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/ › MATLAB Drive

Editor

adc exp 2.m × untitled2.m × untitled.m ×

/MATLAB Drive/untitled.m

```
clc;
clear;
close all;

% Parameters
fs = 100000;
t = 0:1/fs:0.02;
fm = 500;
fc = 10000;
fLO = 14550;
fIF = abs(fLO - fc);

% Message Signal
m = cos(2*pi*fm*t);

% Carrier Signal
c = cos(2*pi*fc*t);

% AM Signal
am = (1 + m).*cos(2*pi*fc*t);

% Local Oscillator
LO = cos(2*pi*fLO*t);

% Frequency Mixing
mixed = am .* LO;

% IF Bandpass Filter
[b,a] = butter(4,[fIF-1500 fIF+1500]/(fs/2),'bandpass');
IF = filter(b,a,mixed);

% Envelope Detection
env = abs(hilbert(IF));

% Remove DC Component
recovered = env - mean(env);

% Display Results
figure('Name','AM Superheterodyne Receiver','NumberTitle','off');
```

